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Collection: *Datasets From Real-Time In-Situ Soil Monitoring for Agriculture 2025*

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ABSTRACT The *Real-time In-situ Soil Monitoring for Agriculture* (RISMA) dataset is a collection of publicly available, high-quality soil volumetric water content (VWC), soil temperature, and meteorological data for agricultural regions in Manitoba, Saskatchewan, and Ontario, Canada. The RISMA network was established beginning in 2011, and data collection continues at the time of publication. Currently, datasets are available for 36 VWC monitoring stations, where sensors are located within annually cropped and perennial pasture sites. Available data varies depending on location but include soil VWC and soil temperature from surface to as deep as 1.5 m, rainfall, air temperature, relative humidity, wind speed, wind direction, and solar radiation. The RISMA stations cover a wide variety of soil types, from clay and clay loams to sandy loams and sand. The data are processed using an automated script which includes a quality control process. This dataset is valuable for researchers working in agriculture, soil science, meteorology, and remote sensing. Data are used to calibrate and validate remote sensing products as well as hydrological, meteorological, and agricultural models. Sites within Manitoba were extensively detailed as core validation sites for NASA's soil moisture active passive (SMAP) satellite.

IEEE SOCIETY/COUNCIL Geoscience and Remote Sensing Society

DATA TYPE/LOCATION Time-series; Saskatchewan, Manitoba, Ontario, Canada

DATA DOI/PID <https://agriculture.canada.ca/en/agricultural-production/soil-and-land/real-time-situ-soil-monitoring-agriculture>

INDEX TERMS Agriculture, calibration, remote sensing, soil moisture, soil water, weather, validation

BACKGROUND

Soil water content data are essential to agriculture given the impact of temporal and spatial variability of root zone water content on crop growth. The data are a component of weather forecasting, drought monitoring and early risk warnings, flood forecasting, crop productivity assessments, and detailing water, energy, and carbon cycles.

The *Real-time In-situ Soil Monitoring for Agriculture* (RISMA) network was established by Agriculture and Agri-Food Canada scientists to provide a direct source of in-situ, near real-time soil water conditions and support efforts to provide remotely-sensed data in hydrologically risk-prone agricultural watersheds [1]. The intention was to verify and improve the performance of algorithms for satellite soil water products [1], [2]. The original stations were established in two major Canadian agricultural regions: the Canadian Prairie region, with a semi-arid, continental climate, and Eastern Ontario with a humid, continental climate. RISMA stations are located in annual and perennial cropping systems to capture the unique microclimate and water flux data associated with Canadian agriculture practices. The stations are distributed so analysis can be done at national, watershed, and field scales, all relevant to agriculture. Open access to the dataset has led to a significant broadening of the user community.

RISMA data have been used extensively in the validation of soil water content estimates from microwave satellites. The Ontario and Manitoba sites are two of 40 international in-situ soil water monitoring sites for the National Aeronautics and Space Administration (NASA) Soil Moisture Active Passive (SMAP) satellite, and recently, the NASA and Indian Space Research Organization (ISRO) NASA-ISRO Synthetic Aperture Radar (NISAR) Mission. These data have been used extensively for global soil water content monitoring products, in development [3], evaluation and improvement of satellite retrieval algorithms [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], and calibration and validation of readings [2], [15], [16], [17], [18], [19], [20], [21], [22], [23]. RISMA, along with other in-situ networks established in agricultural landscapes, continue to be important to calibrate and validate soil water content retrievals for cropland. In addition to validation of SMAP products, RISMA data have been used to test soil water estimates derived from synthetic aperture radar (SAR) satellites including the Canadian Space Agency's RADARSAT [24], [25] and the European Space Agency's (ESA) Sentinel-1 satellite [26]. Additionally, data have been used to validate ESA's Copernicus products including Copernicus Global Land Products "Ground-Based Observations for Validation" (GBOV) [27].

The RISMA data are being applied to large-scale agricultural monitoring products including forecasting crop production as an early warning of regional to global production issues [28], and for Canada's Land Surface model with application to field-scale crop scheduling and disease assessment [29], [30], [31]. The data are also being applied to

the calibration and validation of hydrological modeling for flood forecasting [32], assessing the hydrological budgeting of models in agroecosystems [33], and soil water forecasting [34]. Additionally, the RISMA network has been used to test field equipment such as assessing the performance of Steven's Water Monitoring Systems, Inc. HydraProbe sensors [35], and an ongoing project to test Internet-of-Things (IoT) sensors [36].

RISMA data, along with similar calibration and validation sites from around the globe are included in a database maintained by the International Soil Moisture Network (ISMN) [37]. The ISMN is a collaborative initiative to create a global in-situ soil water database aimed at improving remote-sensing and modelling of soil water. Between 2011 and 2024, there were over 12 000 download requests from 51 countries for RISMA data on ISMN. The requests come from higher educational institutes, research organizations, non-profits, public agencies, and private companies working in agriculture, water, climate, ecosystems, disasters, weather, biodiversity, health, and energy. As of 2021, the ISMN database contained 71 networks and 2842 stations, with data collection beginning as early as 1952 [37]. In 2021, the ISMN hosted 15 stations from North America, 28 from Europe, 3 from Australia, and 19 from Asia [37]. RISMA is the only Canadian network contributing data to the ISMN.

Within the Canadian Prairies, complimentary weather data are available from Environment and Climate Change Canada (ECCC), a Government of Canada department which operates weather stations across Canada. ECCC and the University of Guelph (UG) operated an intensive soil water and weather network near Kenaston, SK from 2007 to 2023 contributing to drought research, cold regions hydrology, and remotely-sensed soil water content [38]. Manitoba Agriculture collects complimentary weather and soil water data in communities across the province to support agriculture, including drought monitoring. RISMA data can support gap-filling in other networks when regional data are missing.

The RISMA network was developed to support data-driven technological advancements such as remote-sensing and model development as opposed to being developed as a widespread monitoring network. RISMA stations are strategically located to capture unique crop microclimates to support remotely-sensing and modelling. The RISMA network is distinguished from other global in-situ soil water networks by several key attributes supporting its research-driven purpose. First, RISMA provides 15-min soil and meteorological data with hourly transmission, enabling near-real-time applications in agricultural monitoring, which surpasses the temporal resolution of many other networks. Second, RISMA employs a multi-depth soil sensor configuration from surface as deep as 1.50 m, with sensors installed in triplicate at each depth to quantify site-level variability and ensure data redundancy—features not uniformly implemented across networks such as The United States Department of Agriculture Soil Climate Analysis Network

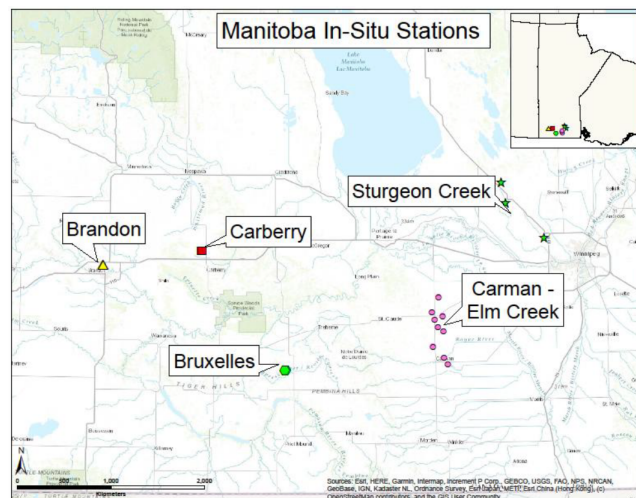


FIG. 1. Map of RISMA stations located in Manitoba.

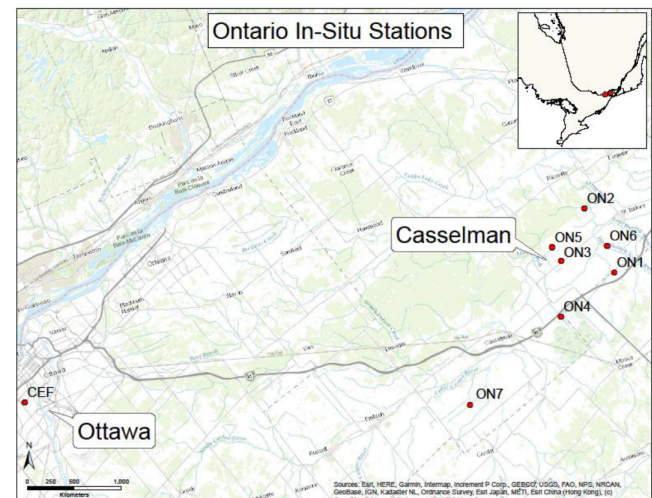


FIG. 2. Map of RISMA stations located in Ontario.

(SCAN) [39] or Cosmic-ray Soil Moisture Observing System (COSMOS) networks [40]. Third, RISMA includes a rigorous calibration protocol, with site-specific calibration models, resulting in higher accuracy ($R^2 \approx 0.95$) compared to factory calibration ($R^2 \approx 0.89$). Fourth, co-located meteorological observations—including precipitation, air temperature, humidity, wind, and solar radiation—are available at many stations, and in some cases complemented by flux towers for energy balance and evapotranspiration estimates. Finally, RISMA stations are situated within actively managed agricultural fields across diverse Canadian ecozones, emphasizing the network’s agri-environmental specificity and relevance to satellite validation and agronomic modeling. These combined characteristics make RISMA a valuable and complementary addition to global networks such as ISMN, where it remains Canada’s only contributing network.

COLLECTION METHODS AND DESIGN

There are data available from 36 RISMA soil water content monitoring stations with datasets located across three diverse Canadian agricultural regions. There are 24 datasets from Manitoba (plus two meteorological stations), 8 from Ontario, and 4 from Saskatchewan (Table I, Supplementary Material 1). Site metadata details and descriptions are available in a document on the repository website [41]. In addition, an interactive map is available by following *Map of monitoring stations* at the provided repository link.

The Manitoban RISMA stations were established beginning in 2011 (Fig. 1, Table I, Supplementary Material 1). The communications, power, and meteorological equipment are located at the edge of annually cropped agricultural fields, with soil volumetric water content (VWC), electrical conductivity (EC), and temperature sensors extending 6 to 30 m out into the field below cropped areas. Nine stations in the Carman/Elm Creek area are located in the Red River Basin (MB 1–9). These sites were selected based on location in a flat terrain, soil texture, soil surveys, and landowner

cooperation to capture diverse soil textures and therefore soil water dynamics, over a targeted region for remote-sensing calibration and validation [35], [42]. Three stations are located in the Sturgeon Creek watershed to capture data from the Assiniboine River Basin and study flood forecasting (MB 10–12) [32]. Additional stations were added for targeted field studies with three near Carberry (MB 13–15) to study irrigation and were decommissioned in 2021. Six stations were installed near Brandon (MB 21–26) with a central weather station (MB 20), and three near Bruxelles, MB (MB 17–19) with a central weather station (MB 16) each set located in one field. These sites were developed to study precision agriculture and greenhouse gas emissions in a slightly rolling terrain.

Seven Ontario stations were established in the Casselman area in the South Nation Watershed beginning in 2011 (ON 1–7), with one additional site near AAFC’s Central Experimental Farm (CEF 1) (Fig. 2, Table I). These sites were selected to capture diverse soil textures in a flat terrain for remote-sensing calibration and validation. The stations are located on the edge of cropped areas, where shallow soil sensors (0.05–0.20 m) are on the edge of the field, and deep soil sensors (0.50–1.00 m) are located 10 m into the field. ON 1 was decommissioned at the request of the landowner in 2015 and replaced with ON 6 in the same year.

The four Saskatchewan stations (SK 1–4) are located in the South Saskatchewan River Basin on pastureland, with the station and soil sensors located within an animal exclusion area within each field (Fig. 3, Table I). These stations were established for remote-sensing calibration and validation to complement additional observation stations operated by ECCC and the University of Guelph that were co-located in a 40 × 40 km area from 2007 to 2023 [38].

At each RISMA location, soil VWC, temperature, and EC, along with meteorological data are collected. Soil sensors (HydraProbe, Stevens Water Monitoring Systems, Inc.) measure soil dielectric permittivity; real permittivity is

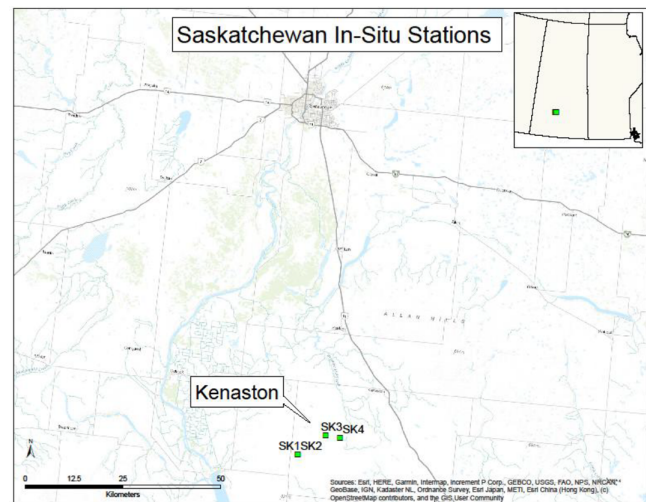


FIG. 3. Map of RISMA stations located in Saskatchewan.

TABLE I. Data Available for Each Station

Network	Stations	Years	Data Category*
Ottawa, ON	CEF 1	2011–2025+	1
Casselman, ON	ON 1–7	2011–2025+	1
Carman-Elm Creek, MB	MB 1–9	2012–2025+	2
Sturgeon Creek, MB	MB 10–12	2014–2025+	2
Carberry, MB	MB 13–15	2015–2021	2
Bruxelles, MB—weather	MB 16	2020–2025+	3
Bruxelles, MB	MB 17–19	2020–2025+	4
Brandon, MB—weather	MB 20	2021–2025+	3
Brandon, MB	MB 21–26	2021–2025+	4
Kenaston, SK	SK 1–4	2013–2025+	2

Note: *1 soil temperature, soil volumetric water content (VWC), soil electrical conductivity (EC) and liquid precipitation; 2 soil temperature, soil VWC, soil EC, liquid precipitation, air temperature, relative humidity, wind speed, wind direction and solar radiation; 3 liquid precipitation, air temperature, relative humidity, wind speed, wind direction and solar radiation; 4 soil temperature, soil VWC, and EC.

converted to VWC using models, imaginary permittivity is converted to salinity, and pore water EC is used to determine temperature.

The sensors are installed with a vertical orientation at surface from 0.00 to 0.05 m and a horizontal orientation in a soil pit, typically at depths of 0.05, 0.20, 0.50 and 1.00 m on a 1 × 1 m vertical wall face (Fig. 4, Supplementary Material 1). Sensors are installed in triplicate approximately 20 cm apart. The redundancy accounts for sensor failure and gives a measure of variability within each site. Data are available for each replicate. Sensors are vertically offset to avoid disturbing downward flow. A soil coring device was used to extract undistributed cores, the hole was cleaned for a flush installation, and the cavity was backfilled with a mixture of soil and bentonite to prevent preferential flow. The soil was then carefully backfilled following the original soil horizonation. For MB 1–9, CEF, and ON 1–7 during

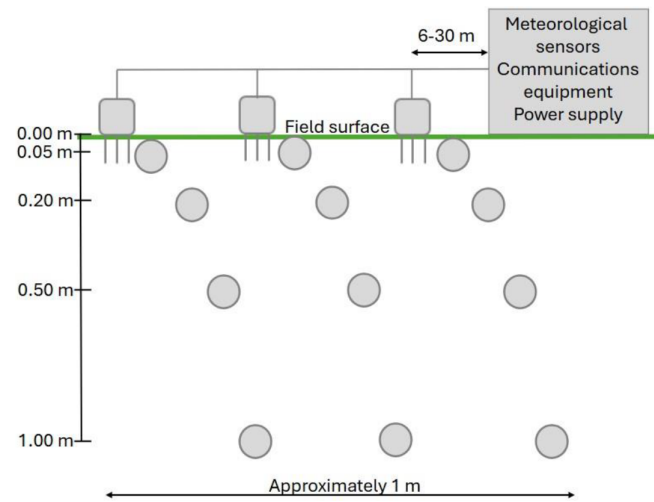


FIG. 4. A schematic of the triplicate installation of RISMA soil sensors. Circles represent horizontal installation into a pit wall, while surface soil sensors are installed vertically.

installation of the sensors, soil samples were collected at the location of each sensor and analyzed in the lab for gravimetric soil water content, bulk density, and soil texture [35], [42].

Meteorological data are collected at each site. Some stations measure soil temperature, soil VWC, soil EC, and liquid precipitation (Data category 1, Table I), while others include variables to estimate evapotranspiration and measure soil temperature, soil VWC, soil EC, liquid precipitation, air temperature, relative humidity, wind speed, wind direction, and solar radiation (Data category 2). Stations MB 17–19 and MB 21–26 are each clustered within one field and each set uses a single weather station to represent the group measuring liquid precipitation, air temperature, relative humidity, wind speed, wind direction, and solar radiation (Data category 3) while the individual stations measure soil temperature, soil VWC, and soil EC (Data category 4). For the clustered stations, the soil sensor stations are a maximum of 525 m from the weather station. SK 4 is co-located with an evaporative flux and meteorological tower operated by ECCC and the University of Saskatchewan [38]. The flux tower measures radiation balance, surface temperature, precipitation, soil heat flux, wind, PAR, snow depth, leaf wetness, barometric pressure, and is equipped with an eddy covariance system.

Instrumentation at the Saskatchewan and Manitoba stations includes: soil VWC, temperature, and EC sensors (HydraProbe, Stevens Water Monitoring Systems, Inc.), tipping bucket rain gauges (TB4, Hydrological Services Pty. Ltd.), anemometers (05103 or 05178A RevD, R.M. Young Company), temperature and relative humidity probes (HMP155A, Vaisala or HC2-S3 or HC-S3-XT, Rotronic) in naturally-ventilated radiation shields (41003-X, R.M. Young Company), pyranometers (LI-200, LI-COR), data loggers (CR1000-XT, Campbell Scientific, Inc.) and 4G or LTE Network (HSPA) modems. The stations are powered by

50–100 W solar panels and one 12-volt 100 AH battery. Installations are set up to capture year-round data with minimal disturbance. From 2011 to 2013 the Manitoban stations were equipped with only soil VWC sensors (SD-12, Stevens Water HydraProbes) and tipping bucket rain gauges.

The instrumentation at the Ontario stations includes soil VWC, temperature, and EC sensors (HydraProbe SD-12, Stevens Water Monitoring Systems, Inc.), tipping bucket rain gauges (RG-1, Adcon), data loggers (CR1000X, Campbell Scientific, Inc.), and 4G modems. The stations are powered by 50 W solar panels and one 12 V 33 AH battery. The CEF station is located 500 m from ECCC's Ottawa CDA meteorological station (Station ID 6105976).

The area above and around each sensor nest is managed to replicate the management of the surrounding field, i.e., as perennial pasture or as the farmer manages their annual crop. Seed, collected from the farmers in Manitoba and provided by AAFC in Ontario, is planted around the installation by hand or small equipment. Weeds are controlled via spray and hand removal. The sites are hand-harvested to coincide with harvest of the main part of the field. In Manitoba, some sites are hand-tilled to replicate tillage practices. In Ontario, to minimize disturbance to the shallow soil sensors, plots are lightly raked. For locations where deep sensors are offset from shallow sensors the deep soil sensors receive the same management the rest of the cropped area. In Saskatchewan, as previously mentioned, the sites are excluded from the rest of the pasture area but are left mostly untouched. Large weeds are removed, but the grasses are not mowed nor are attempts made to match grazing patterns of whatever livestock may be in the pasture.

Weather and soil measurements are collected at 15-min intervals and transmitted hourly via Campbell Scientific Inc., LoggerNet software. The data are stored on AAFC internal servers. Scripted and manual processes are used to calibrate and quality control the data, as detailed below. The data are then displayed in near real-time on the Government of Canada Living Labs Time-Series Portal, accessible through the *Government of Canada Real-Team In-Situ Monitoring for Agriculture* website (agriculture.canada.ca/en/agricultural-production/soil-and-land/real-time-situ-soil-monitoring-agriculture).

VALIDATION AND QUALITY

Quality assurance and quality control (QA/QC) as well as data validation are conducted using field, computational, and manual methods. We have opted to not remove erroneous sensor reading from the dataset, so it is recommended that users complete their own quality control procedures prior to using the data.

Soil sensors are installed in triplicate to ensure minimal disruption to data collection and minimize site disturbance when sensors fail. Apart from the four decommissioned stations, all stations have been continuously operating since installation. Trained technicians manually monitor incoming

data to observe and correct any equipment failures. There are occasional data gaps due to equipment maintenance, damage to stations, sensor failures, or environmental factors. These interruptions are short in duration, ranging from a few days to several weeks. Efforts are made to replace failed sensors as soon as possible to reduce the length of data gaps, especially during the growing season. When sensors fail, shallow soil sensors are replaced as needed while deep sensors are left in place and data recording is terminated. In the occurrence of failure of multiple deep sensors, the station may be re-installed following the process described above. When soil sensors are re-installed, the data are treated as a continuous dataset. No prolonged or systematic data loss has occurred at any station. Beyond the triplicate installation, there is no process to fill gaps in data—data are left as is.

All HydraProbes are individually calibrated for local soil types (Supplementary Material 2). For MB 1–9, site-specific calibration models were developed for each sensor to improve accuracy of soil VWC, detailed in Ojo et al. [35]. These soil sensors are estimated to have $R^2 \approx 0.95$ between observed water content and the square root of the real dielectric constant. This procedure was applied to the Ontario stations, and the results showed that they performed well within the standard calibration values [43]. MB 17–26 were calibrated with similar accuracy. The maximum anticipated calibration error has $R^2 \approx 0.89$ between observed water content and the square root of the real dielectric constant. RSME values were below $0.04 \text{ m}^3 \text{ m}^{-3}$ for calibrated sites and as high as $0.188 \text{ m}^3 \text{ m}^{-3}$ with default calibrations [35]. All sensor measurements are otherwise assumed to be within error tolerances documented for each piece of equipment.

Quality control is applied to incoming data using a script (Supplementary Material 2). The data are graded based on six flags and grade codes are available with the data (Table II, Fig. 5). The Flags are based on physical possibility of readings and recommendations from the Stevens HydraProbe user manual [44]. Flag 1 is activated when soil VWC (also known as water fraction by volume (WFV) ($\text{m}^3 \text{ m}^{-3}$) is out of range, i.e., less than $0.02 \text{ m}^3 \text{ m}^{-3}$ or greater than $0.60 \text{ m}^3 \text{ m}^{-3}$, indicating unrealistic readings. Flag 2 activates if soil VWC ($\text{m}^3 \text{ m}^{-3}$) is more than 10% different from the average among soil sensors, which may indicate an error. Flag 3 activates when soil temperature is below 0°C , indicating frozen soil. Flag 4 activates if the real dielectric constant or temperature corrected soil conductivity are less than 2.4, indicating poor or no contact with soil. Flag 5 activates if the dielectric loss tangent is greater than 1.5, indicating very saline or wet soils which may impact readings [44]. Flag 6 activates if the conductivity reading is greater than 0.2 S m^{-1} , this indicates higher salinity in soils or very dry soils, which can result in reduced accuracy in volumetric water content readings. The HydraProbe will not read EC above 1.5 S m^{-1} [44]. Alert flags for poor quality data can be downloaded with the dataset.

TABLE II. Data Flagging Information for the Risma Network

Flag	ATS Code	Error Message	Error Criteria	Parameter Evaluated	Result
FLAG 1	–2: Unusable	Out of WFV range	<0.02 or $>0.6 \text{ m}^3 \text{ m}^{-3}$	VWC	VWC stays
FLAG 2	–2: Unusable	Out of WFV average range	$>10\%$ different from average	VWC	VWC stays
FLAG 3	3: Ice	Frozen soil	$<0^\circ\text{C}$	Soil temp	VWC stays
FLAG 4	–2: Unusable	Out of RDC range	<2.4	Real dielectric constant	VWC stays
FLAG 5	11: Poor	DLT ≥ 1.5	≥ 1.5	Dielectric loss tangent	VWC stays
FLAG 6	–2: Unusable	Con ≥ 0.2	$\geq 0.2 \text{ S/m}$	Conductivity	VWC stays
Default	31: Good	OK	None	VWC	VWC stays
NA	–1: Unverified	Low battery	NA	Battery voltage	VWC stays
NA	0: Undefined	NA	NA	All others	VWC stays
NA	51: Excellent	NA	NA	Battery voltage	VWC stays

Note: ATS: aquarius time series; WFV: water fraction by volume; RDC: real dielectric constant; VWC: volumetric water content.

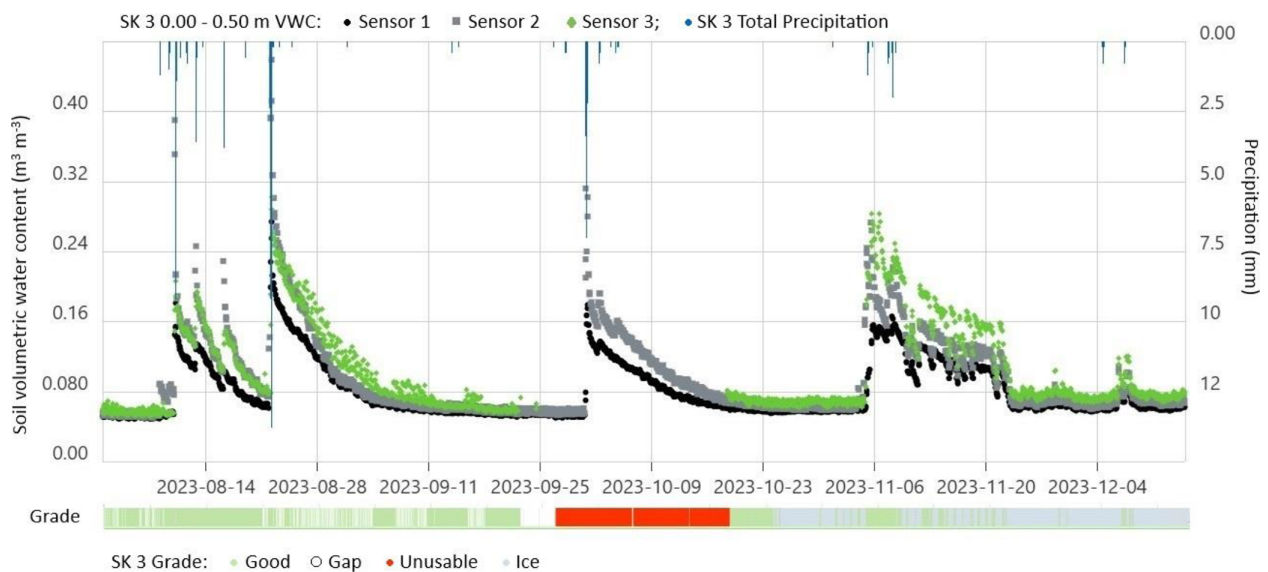


FIG. 5. Example of data from three soil volumetric water content sensors at SK3 from 0.00 to 0.05 m depth with the grade for SK3 0.00–0.05 m sensor 3. This graph highlights variability between soil sensors, data gaps caused by equipment failure, coverage by replicated soil sensors, and responses of the soil water sensors to precipitation during frozen and non-frozen periods.

Dielectric sensors do not work in frozen soil. Soil VWC readings flagged as frozen do not represent soil moisture conditions. This occurs during freeze/thaw cycles and periods of frozen soil (i.e. winter). The different response of the soil VWC readings during frozen periods can be observed in Fig. 5, where flags for frozen conditions at 0.00–0.05 m begin around October 23, 2023 and there is a noticeable difference in response to precipitation events in early November, which can be assumed to be a result of frozen soils. We have elected to leave the frozen soil data in the dataset; however, it is flagged as frozen as part of our automated QA/QC process.

The final step in QA/QC is a manual examination of the data by trained technical staff. Fieldwork that may impact readings is compared with the data, for example disturbance or replacement of sensors (e.g. triggering tipping buckets). Data are visually reviewed for anomalies or poor

readings. At this point erroneous data may be removed. Data on sensors changes, site maintenance, station replacements etc. are extensive and no database is currently available. Although some of this data is captured by the flagging process some may be missed. Information can be requested directly from our authors on these events.

Data accessed through ISMN has additional automated data quality control applied by that organization [37].

RECORDS AND STORAGE

Calibrated and QA/QC RISMA data are available on the Government of Canada Living Labs Time-Series Portal. The datasets are sorted by location. There are currently 36 stations with available data (Table I). An interactive map of the station locations is available on the portal. Interactive charting features can be used to visualize data (e.g. Fig. 5).

TABLE III. Data Available for Download

	Units	Type
Station name	-	Static
Location type	-	Static
Folder	-	Static
Latitude/longitude	WGS 84	Static
Elevation	m	Static
Time stamp	UTC	Static
Description	-	Static
Air temperature	°C	Time series
Precipitation total	mm	Time series
Relative humidity	%	Time series
Wind direction	degree	Time series
Wind velocity min	m s ⁻¹	Time series
Wind velocity max	m s ⁻¹	Time series
Wind velocity	m s ⁻¹	Time series
Soil temperature	°C	Time series
Soil VWC	m ³ m ⁻³	Time series
Soil VWC grade	-	Time series
Battery voltage	V	Time series
Battery voltage grade	-	Time series

Note: *VWC: volumetric water content.

The Portal can be used to export selected parameters and time periods in CSV, Excel, and JSON formats (Table III). The data export contains station numbers and data with units and grades (Table III). Data quality grades can be downloaded with the datasets.

The data are stored internally on a Government of Canada server. Unedited raw data from the data logger are stored, as are QA/QC data. Data for some stations are available prior to 2013 by direct request.

Additionally, the data are available from ISMN, a repository funded by the European Space Agency to store in-situ soil water measurements [37]. Work is ongoing to have near real-time data available at that location.

INSIGHTS AND NOTES

This dataset contributes to research in the domains of remote-sensing, agricultural and hydrological modeling, cold-regions challenges, and soil processes.

SOURCE CODE AND SCRIPTS

The RISMA calibration and QA/QC script applies the calibration equations to calculate soil VWC, grade data, based on the parameters described above, and transfers the data to the repository. The script was developed to apply site-specific calibration and apply quality control to the data. The script will be published in association with this article (Supplementary Material 2).

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Prepared visualizations: K.M., K.G., and A.H.

All authors reviewed the manuscript.

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